

claude-windows / 6e9d0958-a2ec-4a8e-bb2d-6cce9833c527

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6e9d0958-a2ec-4a8e-bb2d-6cce9833c527\subagents\agent-a1ead067841180bea.jsonl`  
- SHA-256: `62bbd4948c70a8e65ddaf5a852d6007e479e4902e228ef63f5d3efe9ed9f29b2`  
- Source modified: `2026-07-03T16:40:49+00:00`  
- Imported at: `2026-07-05T16:48:26+00:00`  
- project: `subagents`  
- session_id: `6e9d0958-a2ec-4a8e-bb2d-6cce9833c527`
```

Transcript

1. user (2026-07-03T16:40:11.675Z)

You are a code-review finder agent. Review target: PR #451 of the badminton-highlight-indexer repo, checked out at C:/Users/avidu/Projects/khelsutra-guru/badminton-highlight-indexer on branch codex/alpha-readiness-plan (HEAD cd07339). Get the diff with: git diff origin/master..HEAD. Docs-only PR: new docs/ALPHA_LAUNCH_READINESS.md plus index additions.

ANGLE ? simplification. Flag unnecessary complexity the new doc adds that will cost maintenance: sections that restate each other verbatim (e.g., §8 "next PR sequence" vs §7 tracker ? do they agree, and is one derivable from the other, risking drift?), §10 definition-of-done items vs §7 rows (do they map 1:1 or diverge?), redundant status legends, or content that belongs in an existing doc instead. Only flag where you can name a concrete divergence-or-drift cost and the simpler form (e.g., "§8 step 3 orders hosted-alpha before baselines while §7 A4 has no dependency on A1 ? derive §8 from §7 or delete it").

Return up to 6 candidate findings as a JSON array of {file, line, summary, failure_scenario}. Return [] if none. Your final message must be only the JSON array.

2. assistant (2026-07-03T16:40:13.474Z)

I'll analyze the PR #451 for simplification issues, focusing on unnecessary complexity, redundancy, and maintenance costs.

3. user (2026-07-03T16:40:14.319Z)

```
diff --git a/docs/ALPHA_LAUNCH_READINESS.md b/docs/ALPHA_LAUNCH_READINESS.md  
new file mode 100644  
index 0000000..4b3d49d  
--- /dev/null  
+++ b/docs/ALPHA_LAUNCH_READINESS.md  
@@ -0,0 +1,171 @@  
+# Alpha Launch Readiness Plan  
+  
+> **Status:** `DRAFT` - cross-repo execution tracker; **Owner:** @avidullu; **Created:**  
2026-07-03; **Last updated:** 2026-07-03  
+> **Lifecycle:** `DRAFT -> IN PROGRESS -> DONE -> archived`  
+> **Tracking anchors:** Section 7 is the progress tracker and source of truth; indexed in  
`docs/README.md` and `docs/DOC_STATUS.md`.  
+> **Relation to existing docs:** consolidates the launch-relevant pending items from  
`NEXT_STEPS.md`, `ROADMAP.md`, `data_pipeline/GOLDEN_VIDEOS.md`, `R2_EVAL_METRIC_DESIGN.md`,  
`COMPUTE_DECOUPLED_SERVING/`, and the sibling `khelsutra` Studio docs.  
+> **Honesty note:** `[verified]` means checked against the local synced repos on 2026-07-03.  
`[analysis]` means derived from a local command run on that snapshot. This is not legal,  
financial, or launch approval.  
+  
+---  
+
```

```

+## 0. TL;DR
+
+The corpus is now large enough to move from "grow labels first" into alpha-readiness work, but
the 15-video headline is not itself a launch gate pass. `[verified]` The golden manifest has 15
badminton videos; `[analysis]` the local manifest needed path rebasing before manifest-driven
commands ran; `[analysis]` only 6 of 15 feature JSONs had the full fusion schema; `[analysis]`
the n=15 quality baseline is stale and must be intentionally refreshed before it can gate
anything.
+
+The first controlled alpha should be a hosted, authenticated product loop: upload a video,
process it, select rallies, compile, and download a reel. The owned-model/commercial rally-
detection claim remains a separate NO-GO until the ship-gate target is ratified and the WASB
checkpoint provenance is resolved.
+
+## 1. Problem & goal
+
+The previous launch blockers were spread across roadmap docs and mixed together three different
questions:
+
+- **Can an invited tester use the product without SSH?**
+- **Can the 15-video golden corpus now support honest quality gates?**
+- **Can we claim or ship an owned/commercial rally detector?**
+
+This doc separates those into executable workstreams. The goal is to reach an invite-only alpha
where the product is usable end-to-end, the quality claims are honest and reproducible, and the
remaining model/legal gates are explicit rather than accidentally implied by launch activity.
+
+## 2. Decisions locked
+
+| # | Decision | Source / date | Implication |
+|---|---|---|---|
+| D1 | **Product alpha and owned-model launch are separate decisions.** | `NEXT_STEPS.md` NO-GO
banner, 2026-06-30; verified 2026-07-03 | We can launch a controlled hosted product loop without
claiming the owned model is commercially cleared. |
+| D2 | **First alpha must be page-driven, not SSH/CLI-driven.** | `NEXT_STEPS.md` P0 UI-
driven/shareable item; verified 2026-07-03 | Hosted service, edge auth, upload, job polling,
compile, and download are launch-critical. |
+| D3 | **The 15-video corpus is now a measurement asset, not a free launch pass.** |
`data_pipeline/GOLDEN_VIDEOS.md`; local eval runs, 2026-07-03 | Rebase/portability, feature
completeness, baseline refresh, and gate ratification come before quality claims. |
+| D4 | **Do not promote served Gate A by default from the current n=15 evidence.** |
`backend.eval.serve_contrast` run, 2026-07-03 | Proposal-side lift does not survive served-
windowing safety; keep it as an opt-in/analysis lever. |
+| D5 | **P10/P11 corpus promotion and train/re-eval are alpha-plus unless owner reopens D13.**
| sibling `khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`; verified 2026-07-03 | First alpha can use
manual/offline corpus promotion; automated flywheel follows after serving and gates are stable.
|
+
+## 3. Verified snapshot, 2026-07-03
+
+All five Git checkouts were fast-forwarded before analysis:
+
+| Repo | Branch | Synced HEAD |
+|---|---|---|
+| `badminton-highlight-indexer` | `master` | `e4914d1` |
+| `khelsutra` | `master` | `b0f70a5` |
+| `rally-annotator` | `main` | `ef9e088` |
+| `sports-data-collector` | `master` | `bfc014f` |
+| `sports-obsreport` | `main` | `a0c7252` |
+
+Corpus/eval facts:
+
+- `[verified]` `docs/data_pipeline/GOLDEN_VIDEOS.md` says `output/human_lovo_manifest.json` is
the ingested golden corpus at `n = 15`.
+- `[analysis]` The local manifest had stale absolute paths. A temporary rebased manifest found

```

all 15 trajectories and labels in this workspace.

```

+- `[analysis]` There are 15 `output/*_golden_features.json` files, but only 6 carry full
shuttle/person/audio candidate features. The newer 9 are windowing-only schema and are skipped
by fusion/person eval.
+- `[analysis]` `python -m backend.eval.fusion_compare --features-dir output` loaded only 6
videos and measured person-fusion `Delta F1 = -0.021`; not promotable.
+- `[analysis]` `python -m backend.eval.serve_contrast --manifest <rebased-n15>` measured
proposal Gate A mean `Delta F1 +0.0287`, but served mean `Delta F1 -0.0328`; not served-safe.
+- `[analysis]` `python -m training.gen0.harness --manifest <rebased-n15> --floor
eval_baselines/heuristic_lovo_n6.json` produced learned LOVO mean F1 `0.551`, `ship=False`;
blockers remain stale n=6 floor/sign test, uncalibrated ship target, and WASB C3 provenance.
+- `[analysis]` `python scripts/nightly_regression.py --manifest <rebased-n15> --features-dir
output --no-report` reported default/milestone tolF1 `0.3636` vs baseline-off tolF1 `0.1382`,
but marked the frozen baseline stale due to config/corpus changes.
+- `[analysis]` `python -m backend.eval.ablation --features-dir output --granularity group`
currently fails at import due to a circular import between `backend/eval/ablation.py` and
`backend/eval/ablation_pdf.py`.
+
+### 4. Launch strategy
+
+### 4.1 Alpha lane: make the product usable
+
+The alpha lane is successful when an invited tester can open the app, authenticate, upload a
video, watch the job progress, pick rallies, compile a reel, and download it. This lane should
avoid model-quality promises beyond the measured state and should default to the most reliable
serving path.
+
+### 4.2 Quality lane: make the 15-video corpus gate-ready
+
+The quality lane is successful when the 15-video corpus is portable, reproducible, feature-
complete, and backed by refreshed baselines. Only then should R2 tolF1, Gen-0/TemporalMaxer, or
other model/promotion decisions be ratified.
+
+### 4.3 Flywheel lane: make tester feedback useful
+
+The flywheel lane is successful when user corrections can flow into the collector/vault path
with consent/provenance and a visible re-eval delta. This is valuable for alpha learning, but it
should not block the first controlled alpha if manual promotion can bridge the gap.
+
+## 5. Risks and guardrails
+
+| Risk | Why it matters | Guardrail |
+|---|---|---|
+| Launch activity implies model launch approval | The engine docs explicitly say rally-
detection product launch is NO-GO until ship target and WASB C3 clear | Keep all alpha copy
scoped to invite-only processing, not owned-model claims |
+| Stale local manifests create false confidence | Manifest-driven tools skipped or failed until
paths were rebased | Make the corpus manifest portable and add a path/fixture smoke check |
+| 15-video corpus is only partially feature-complete | Fusion/person/audio eval is still n=6
today | Regenerate or upconvert full-fusion features for all 15 before fusion decisions |
+| Baseline refresh accidentally rubber-stamps a regression | Nightly runner correctly reported
stale baselines | Refresh baselines only after reviewer sign-off and record the command/output |
+| P10/P11 expands scope before hosted alpha is usable | Corpus flywheel is valuable but can
consume the alpha launch window | Keep P10/P11 behind explicit owner reopen unless alpha lane is
already stable |
+| WASB checkpoint provenance remains unresolved | Blocks commercial sharing of a trained model
regardless of F1 | Treat C3 as a hard model-release gate; do not hide it inside product launch
tasks |
+
+### 6. Honest limits
+
+Completing this doc's alpha lane does not prove the owned model is ready, that WASB weights
are commercially clean, that the product can serve unbounded public traffic, that multi-sport is
ready, or that every tester correction automatically improves the model.
+

```

+Completing the quality lane does ****not**** by itself deploy a service. It only makes the measurement surface honest enough for a launch decision.

+

7. Deliverables & progress tracker

+

+Legend: TODO = not started, IN PROGRESS = actively being worked, DONE = shipped/verified, GATED = blocked on owner/legal/platform decision. One PR per row unless the row explicitly names a cross-repo batch.

+

ID	Deliverable	Repo / owner	Depends on	Gated?	Status	PR	
A0	Create this alpha readiness tracker and index it in docs	`badminton-highlight-indexer`	-	No	IN PROGRESS	#451	
A1	Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check, edge auth ON, upload -> process -> jobs -> compile -> download verified	`khelsutra` + `badminton-highlight-indexer`	A0	Owner spend / CF config	TODO	-	
A2	Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke check that fails loudly when labels or trajectories are missing	`badminton-highlight-indexer`	A0	No	TODO	-	
A3	Promote the 15-video source-video/MD5 records into the collector/vault path; reconcile collector's six-video source manifest with the 15-video eval corpus	`sports-data-collector` + vault	A2	Owner upload/storage scope	TODO	-	
A4	Refresh the n=15 nightly regression baseline intentionally and record the exact command/output; keep stale-baseline warnings until reviewed	`badminton-highlight-indexer`	A2	Reviewer sign-off	TODO	-	
A5	Recompute the heuristic n=15 floor and replace the stale `heuristic_lovo_n6` floor for future Gen-0 comparisons	`badminton-highlight-indexer`	A2	No	TODO	-	
A6	Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation on the n=15 corpus	`badminton-highlight-indexer`	A2, A4	No	TODO	-	
A7	Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so fusion/person/audio analyses run over all 15	`badminton-highlight-indexer`	A2	GPU/data availability	TODO	-	
A8	Re-run `fusion_compare`, group ablation, and served contrast after A7; explicitly promote/reject person fusion, Gate A served-default, and any other R-series default flips	`badminton-highlight-indexer`	A6, A7	Reviewer sign-off	TODO	-	
A9	Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce a non-serving artifact bundle and gate verdict	`badminton-highlight-indexer`	A5, A8	Owner approval for M0.4 scope	TODO	-	
A10	Decide and record the alpha ship-gate target: metric, corpus slice, minimum threshold, significance rule, and what claims are allowed	Owner + `badminton-highlight-indexer`	A4, A5, A8	Owner decision	GATED	-	
A11	Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred, document "no commercial model sharing" copy guardrail	Owner/legal	A10	Legal/provenance	GATED	-	
A12	Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector promotion, `train reeval` job, and delta-F1 surface	`khelsutra` + `badminton-highlight-indexer`	collector	A1, A3	Owner D13 reopen	GATED	-
A13	Alpha launch review: runbook, privacy/consent copy, known limits, rollback/disable steps, and owner GO/NO-GO	Cross-repo	A1, A4, A10, A11	Owner launch decision	TODO	-	

+

8. Concrete next PR sequence

+

1. ****Docs PR:**** land this tracker.
2. ****Corpus portability PR:**** rebase/portable manifest tooling plus missing-file smoke test.
3. ****Hosted alpha PR/batch:**** health/auth/upload/process/job/compile/download verification and runbook updates.
4. ****Baseline PR:**** intentional n=15 nightly baseline refresh and heuristic floor refresh.
5. ****Ablation/fusion PRs:**** fix ablation import, complete 15-video features, rerun promotion/rejection decisions.
6. ****Launch decision PR/docs update:**** record ship-gate target, C3 status, alpha copy limits, and launch checklist.

+

9. Open questions

+

1. ****Owner:**** Is the first alpha allowed to use manual/offline corpus promotion, or should

```

P10/P11 be reopened before any tester wave?
+2. **Owner/platform:** Which hosted compute target is the next stand-up: Cloud Run GPU, GCE L4 VM, or another approved GPU host?
+3. **Owner/legal:** What is the acceptable alpha wording while WASB C3 remains unresolved?
+4. **Engineering:** Should the portable corpus manifest live as a generated artifact, a checked-in redacted manifest, or both?
+5. **Engineering:** Should n=15 feature regeneration target full fusion schema for every clip, or should the eval loaders support mixed schemas with explicit capability tags?
+
+## 10. Definition of done
+
+- [ ] Invited tester can complete the hosted record/upload -> reel download CUJ without SSH.
+- [ ] Alpha service is gated behind the chosen auth posture and has a documented disable/rollback path.
+- [ ] n=15 corpus is portable enough that manifest-driven quality commands do not rely on stale local absolute paths.
+- [ ] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
+- [ ] R2/tolF1 gate is promoted or rejected with a recorded ablation result.
+- [ ] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected on the 15-video corpus.
+- [ ] Ship-gate target and allowed alpha claims are recorded.
+- [ ] WASB C3 status is recorded and reflected in alpha messaging.
+- [ ] P10/P11 are either reopened with a committed sequence or explicitly deferred for alpha-plus.
+- [ ] Owner records GO/NO-GO for the first alpha wave.
+
+## 11. References
+
+Internal docs:
+
+- `docs/NEXT_STEPS.md`
+- `docs/ROADMAP.md`
+- `docs/data_pipeline/GOLDEN_VIDEOS.md`
+- `docs/R2_EVAL_METRIC_DESIGN.md`
+- `docs/RALLY_DETECTION_FIXES_PLAN.md`
+- `docs/COMPUTE_DECOUPLED_SERVING/README.md`
+- sibling `khelsutra/NEXT_STEPS.md`
+- sibling `khelsutra/docs/STUDIO_MEDIA_LIFECYCLE.md`
+- sibling `khelsutra/docs/PER_RALLY_SELECTION.md`
+- sibling `sports-data-collector/data/golden_source_videos.json`
+
+Local verification commands used for the 2026-07-03 snapshot:
+
+```powershell
+python -m backend.eval.fusion_compare --features-dir output
+python -m backend.eval.serve_contrast --manifest <rebased-n15-manifest>
+python -m training.gen0.harness --manifest <rebased-n15-manifest> --floor eval_baselines\heuristic_lovo_n6.json --version analysis-n15 --out <temp-out>
+python scripts\nightly_regression.py --manifest <rebased-n15-manifest> --features-dir output --no-report
+python -m backend.eval.ablation --features-dir output --granularity group
+```
+
+---
+
+#### Changelog
+
+- 2026-07-03 - Created from a synced five-repo review of `NEXT_STEPS`, `ROADMAP`, golden-corpus docs, local eval artifacts, and the current 15-video corpus state.
diff --git a/docs/DOC_STATUS.md b/docs/DOC_STATUS.md
index 34411fd..2541672 100644
--- a/docs/DOC_STATUS.md
+++ b/docs/DOC_STATUS.md
@@ -30,6 +30,7 @@
@@ Before a foundational/tracked doc moves to `docs/archives/`:
### 3a. Tracked projects ? lifecycle docs (archive when DONE; each doc's $7 is the completeness

```

```

source-of-truth)
| Doc | Status | Archive-ready? | Next action |
|---|---|---|---|
+| `ALPHA_LAUNCH_READINESS.md` | **DRAFT** ? cross-repo alpha readiness tracker created
2026-07-03 after the golden corpus reached n=15; separates hosted product-alpha work from owned-
model/commercial rally-detection gates | No | Execute $7 in order: corpus portability + n=15
baselines, hosted alpha serving, ablation/fusion reruns, then launch GO/NO-GO |
| `COMPUTE_DECOUPLED_SERVING/` | **IN PROGRESS** ? M0 approved + **M1?M9 LARGELY SHIPPED**
(#279?#290; M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF); Gen-0 weights
pushed (`weights/0.1.0-l4/`) | No | M11 flywheel + M12 gdrive-api (owner residual:
counsel/prices/OAuth/real-run within the D17 cap) |
| `RALLY_QUALITY_RESEARCH.md` | Foundation, **tracked & ACTIVE** ($7) | No | Drives the
R-series; complete when $7.1 single quality number is attributed |
| `R2_EVAL_METRIC_DESIGN.md` | IMPLEMENTED (augmenting) | No (gates R4 + dual-eval; ablation
pending) | Promote-or-reject tolF1 as the gate (ablation) ? then archive with the R-cluster |
diff --git a/docs/NEXT_STEPS.md b/docs/NEXT_STEPS.md
index f186ff9..a9f35a8 100644
--- a/docs/NEXT_STEPS.md
+++ b/docs/NEXT_STEPS.md
@@ -9,7 +9,7 @@

```

```

## ? Prioritized pending ? all tracks
**P0 ? highest leverage / unblocks the most**
-0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings
lying around ? highlights, from a page, no SSH". ** Today the cloud flow is **SSH/CLI-only**
(`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it. ** Needs (a separate khelsutra `web/`
SPA + deploy track, NOT an engine seam): ** (i) ** deploy the engine as a reachable **service**
(M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), ** (ii) ** the **SPA
ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll
`/api/jobs` ? reel; ZERO engine code), ** (iii) ** flip **M9 edge-auth ON** (merged #290, default-
OFF) to gate alpha tenants, ** (iv) ** a **compute PICKER in the UI** ("your machine vs our
cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full record:
[[ui-driven-cloud-serving-gap]].
+0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings
lying around ? highlights, from a page, no SSH". ** Today the cloud flow is **SSH/CLI-only**
(`docs/CLOUD_GPU_DRYRUN_GUIDE.md`); **M12 does NOT fix it. ** Needs (a separate khelsutra `web/`
SPA + deploy track, NOT an engine seam): ** (i) ** deploy the engine as a reachable **service**
(M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), ** (ii) ** the **SPA
ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll
`/api/jobs` ? reel; ZERO engine code), ** (iii) ** flip **M9 edge-auth ON** (merged #290, default-
OFF) to gate alpha tenants, ** (iv) ** a **compute PICKER in the UI** ("your machine vs our
cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full record:
[[ui-driven-cloud-serving-gap]]. **Concrete execution tracker after the n=15 corpus review:**
`docs/ALPHA_LAUNCH_READINESS.md` $7.
0b. **? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable batteries-
included app ? `docs/PACKAGING_PLAN.md`. ** The LIGHT-tier engine is bundle-ready: `pip install`
? `indexer --app` serves the **bundled SPA** single-origin, **torch-free**, with app-home state
/ ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b,
P2-launcher/P2a/P2d/P2e). **? Next (fresh focused pushes): ** (1) ** P3 **GPU validation +
screencast** (L4, ~$0.50, DELETE-after, $100 cap ? `[[gcp-vm-always-delete]]` ? native-WASB
0-rally tuned-config blocker), ** (2) ** P2b-install (uv/script + truly-local default config),
** (3) ** P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then P3-cloud
(Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-
session-packaging-app]]`.
1. ? **COMPUTE_DECOUPLED_SERVING` M0?M9 LARGELY SHIPPED** (2026-06-21, ADR-014 ratified).
M1?M4 (#279/#280/#282/#283), **M5 #286, M6 #287, M8 #288, M9-auth #290** all merged (default-
OFF); the **real M7 L4 run** validated the worker (22 rallies, VM deleted, $0). All $8 gating Qs
locked (D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel ToS/DPA
(M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF
team-domain/AUD + ingress-lock; real cloud verification authorized within a **$100/?10k cap**
(D17).
2. **[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3 matches/sport (TT
first). The single highest-leverage move; the quality track *and* full-corpus training feed on
it.
diff --git a/docs/README.md b/docs/README.md

```

```

index 0df1a69..58be5ff 100644
--- a/docs/README.md
+++ b/docs/README.md
@@ -15,6 +15,7 @@
- [DOC_STATUS.md](DOC_STATUS.md) ? **? living Documentation Status & Archival Register**: every
foundational/tracked doc's lifecycle + archive-readiness, the **archival due-diligence
checklist** (*honestly update ? then archive*), and the open doc-hygiene findings. Review here;
update rows as docs complete.

## Product / platform plans
+- [ALPHA_LAUNCH_READINESS.md](ALPHA_LAUNCH_READINESS.md) ? **? Active tracked project (`DRAFT`,
created 2026-07-03):** cross-repo alpha readiness tracker after the golden corpus reached 15
videos. Separates the hosted product-alpha lane (shareable upload?process?pick?reel) from the
owned-model/commercial rally-detection gate (ship target + WASB-C3 still unresolved). S7 is the
source of truth for the concrete next PR sequence: corpus portability, n=15 baseline refresh,
ablation/fusion reruns, hosted serving, alpha copy/launch review.
- [COMPUTE_DECOUPLED_SERVING/](COMPUTE_DECOUPLED_SERVING/README.md) ? **? Active tracked
project (2026-06-21, `DRAFT` owner-gated):** the productionization successor to
DECOUPLED_COMPUTE (its deferred M3/M4). **One pure core (DETECT?WINDOW?STITCH) that never
branches on deployment profile**, with a `ComputeBackend` (where it runs) + `StorageBackend`
(what bytes) selected by **one startup config**. Ships **truly-local** (local/USB + local GPU,
in-process stitch, zero cloud) + **cloud-serving** (upload<2GB / gdrive / bucket ? a **Cloud Run
GPU job runs detect AND stitch**, metered into a per-job cost ledger) + **cpu-mock** (CI).
Includes [TESTING_STRATEGY.md](COMPUTE_DECOUPLED_SERVING/TESTING_STRATEGY.md) (profile-PARITY
proof for $0), [architecture.svg](COMPUTE_DECOUPLED_SERVING/architecture.svg), a
[runlogs/](COMPUTE_DECOUPLED_SERVING/runlogs/) posterity dir, and **ADR-014** ([lineage
ADR-009?010?011?012?013?014](archives/decisions/DECISIONS.md)). S7 M0?M10 tracker is the source
of truth.
- [DECOUPLED_COMPUTE/](archives/past_projects/DECOUPLED_COMPUTE/README.md) ? ? **DELIVERED +
ARCHIVED (2026-06-21).** **Lifted the rally pipeline onto a **rented cloud-native GPU** doing
bucket-driven `train` + `infer`: storage/compute seams (PRs #246?#259) + the **M3.5 native WASB
runner**; **M0?M2 + M3.5 proven on a GCP L4** (M2 ? `weights/0.1.0-l4/`). Delivered on **GCP** ?
DigitalOcean dropped for compute ([ADR-013](archives/decisions/DECISIONS.md)). **M3/M4 (serving
endpoint + per-video billing + scale-to-zero) deferred ? the "Cloud Run + GPU serving"
subproject** ([ADR-012](archives/decisions/DECISIONS.md)). Snapshot under
`docs/archives/past_projects/DECOUPLED_COMPUTE/` (incl. `DEPLOY_REPORT_GCP_2026-06-20.md`).
- [PACKAGING_PLAN.md](PACKAGING_PLAN.md) ? **? Active tracked project (`IN PROGRESS`, 11 PRs
merged 2026-06-21/22):** ship the engine as a **downloadable batteries-included Python app** ?
`pip install` ? `indexer --app` boots a local webserver and opens the **bundled KhelSutra SPA**
single-origin, **torch-free** (LIGHT tier), cross-platform, no native component (decision
D10/O1). Built ON the decoupled-compute seam. **Supersedes the `DESKTOP_APP_PLAN.md`
narrative.** S7 tracker = source of truth (P0a?e ?, P1a/b ?, P2-launcher/P2a/P2d/P2e ?). **?
Next:** P3 GPU validation + screencast ? P2b-install ? P2c wizard (cross-repo) ? P4
(ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-
app]]`.

```

4. assistant (2026-07-03T16:40:16.207Z)

Now let me check the existing docs structure to understand what's being consolidated:

5. user (2026-07-03T16:40:16.879Z)

```

1      # Next steps (live) ? refreshed 2026-06-21
2
3      > **How to read:** this top section is the **current cross-track state + prioritized
pending list** (refreshed 2026-06-21, replacing the stale 2026-06-13 header). The detailed
**owned-model & rally-quality track** plan (2026-06-13) is **preserved below, still valid** ? it
is now *one of several* active tracks. Live blow-by-blow: memory `current-state.md`. Doc
lifecycle/hygiene: `docs/DOC_STATUS.md`.
4
5      ## ? Where we are (2026-06-21)
6      **DECOUPLED_COMPUTE delivered on GCP + archived** (#270); **compute = GCP only**
(ADR-013, DO dropped). **`COMPUTE_DECOUPLED_SERVING`** subproject spun up (#272, **ADR-014**) ?
one pure core + local/cloud profiles; **M0?M9 LARGELY SHIPPED** (ADR-014 ratified 2026-06-21;
M1?M4 #279?#283, M5 #286, M6 #287, M8 #288, M9-auth #290 all merged, default-OFF). **Per-video

```

workspace P0** shipped (#264); **doc-status register + archival due-diligence** in place (#273). The **khelsutra site + per-rally UI** ship in parallel (sibling track). The **rally-quality / owned-model** track (below) is the core ML work, gated on **growing the golden corpus**.

7
8 > **? LAUNCH DECISION ? NO-GO (owner, 2026-06-30).** The rally-detection product is **not** cleared for an external/commercial launch; we stay in **development**. This is the expected default for the dev phase. Two unblockers gate a future GO: **(1)** set + meet the **ship-gate target (#172)** ? currently unset, needs corpus growth (and the R2 toF1 ablation, item 8); **(2)** resolve the **WASB shuttle-detector weight provenance** (commercial-clean licensing). P2b (#396) was a do-no-harm quality increment, **not** a launch gate ? merging dev PRs ? launching. Re-decide only when both unblockers clear.

9
10 ## ? Prioritized pending ? all tracks
11 **P0 ? highest leverage / unblocks the most**
12 0. **? [owner-priority, 2026-06-21] MAKE IT UI-DRIVEN + ALPHA-SHAREABLE ? "a few recordings lying around ? highlights, from a page, no SSH".** Today the cloud flow is **SSH/CLI-only** (`docs/CLOUD\_GPU\_DRYRUN\_GUIDE.md`); **M12 does NOT fix it.** Needs (a separate khelsutra `web/` SPA + deploy track, NOT an engine seam): **(i)** deploy the engine as a reachable **service** (M7 ? Cloud Run service, ? owner GCP spend for a *persistent* endpoint), **(ii)** the **SPA ingest/progress screen** (khelsutra `web/` B-P2: enter/upload video ? `/api/process` ? poll `/api/jobs` ? reel; ZERO engine code), **(iii)** flip **M9 edge-auth ON** (merged #290, default-OFF) to gate alpha tenants, **(iv)** a **compute PICKER in the UI** ("your machine vs our cloud", D13/D15). The engine seams make it *possible*; the SPA makes it *usable*. Full record: [[ui-driven-cloud-serving-gap]]. **Concrete execution tracker after the n=15 corpus review:** `docs/ALPHA\_LAUNCH\_READINESS.md` §7.

13 0b. **? [packaging track, IN PROGRESS ? 11 PRs merged 2026-06-21/22] Downloadable batteries-included app ? `docs/PACKAGING\_PLAN.md`.** The LIGHT-tier engine is bundle-ready: `pip install` ? `indexer --app` serves the **bundled SPA** single-origin, **torch-free**, with app-home state / ffmpeg resolver / DNS-rebinding guard / single-instance lock / safe DB upgrades (P0a?e, P1a/b, P2-launcher/P2a/P2d/P2e). **? Next (fresh focused pushes):** (1) P3 **GPU validation + screencast** (L4, ~\$0.50, DELETE-after, \$100 cap ? `[[gcp-vm-always-delete]]` ? native-WASB 0-rally tuned-config blocker), (2) P2b-install (uv/script + truly-local default config), (3) P2c first-run wizard (cross-repo khelsutra SPA from `/api/capabilities`), then P3-cloud (Cloud Run) + P4 (ONNX/train/annotate = north star). Memory `[[packaging-app-plan]]` / `[[next-session-packaging-app]]`.

14 1. ? **`COMPUTE\_DECOUPLED\_SERVING` M0?M9 LARGELY SHIPPED** (2026-06-21, ADR-014 ratified). M1?M4 (#279/#280/#282/#283), **M5 #286, M6 #287, M8 #288, M9-auth #290** all merged (default-OFF); the **real M7 L4 run** validated the worker (22 rallies, VM deleted, \$0). All \$8 gating Qs locked (D7?D17). **Owner residual** (not blocking Claude's default-OFF code): counsel ToS/DPA (M9-consent live + M11 live), real GCP prices+FX (M8 calibrate), Google OAuth app (M12 real), CF team-domain/AUD + ingress-lock; real cloud verification authorized within a **\$100/?10k cap** (D17).

15 2. **[owner, ongoing] Grow the golden corpus** ? multi-court badminton + ?3 matches/sport (TT first). The single highest-leverage move; the quality track *and* full-corpus training feed on it.

16 3. **[owner + Claude . GPU] Full-corpus training (n?5)** ? a real `weights/<ver>/` (the `0.1.0-l4` bundle is n=2 plumbing-only).

17
18 **P1 ? near-term, mostly Claude-doable (default-OFF, green-gated)**
19 3b. **[Claude] Five-repo audit health remediation ? ? COMPLETE (2026-07-02).** All waves (P0?P28) landed and the last residuals were closed or filed as issues; the umbrella tracker was archived to `docs/archives/past\_projects/AUDIT\_REPORT\_khelsutra-guru\_2026-07-02-COMPLETED-2026-07-02.md`. Remaining owner-gated/future work lives as GitHub issues: #443 (CORS default), #444 (corpus Phase-2 upload), #445 (GCS eval plumbing), plus #301/#327/#332/#384.

20 4. **[Claude] `COMPUTE\_DECOUPLED\_SERVING` batch ? REMAINING: M11 + M12** (M5/M6/M8/M9-auth ? merged this session). **M11** = `backend/flywheel.py` capture?store?shadow-eval?goldenize plumbing reusing `workspace.py::for\_video` + `backend/eval/experiment.compare` + `backend/eval/promotion.py::promote\_if\_served\_safe`, behind `flywheel.\*`=OFF + a faked `consent\_attested` (? privacy-sensitive ? give it fresh focus). **M12** = `backend/storage/gdrive\_api.py` `GDriveApiStorage(StorageBackend)` + a `gdrive-api://` scheme (regex already allows the hyphen; add to `\_KNOWN` + `\_backend\_class` + the StorageConfig Literal) against an INJECTED fake Drive (mirror `gcs.py` DI); keep the mount-only `gdrive://` untouched. ? **M9-consent cols = a new v7 migration** (M8 took v6). M9-consent ToS/DPA text = owner/counsel.

21 5. ****[Claude] Per-video workspace P1?P6**** (P1 v6 DB migration ? P2 worker ? P3 per-video routing of `detector/`+`tmp/` ***[the `output/chunks\_` de-hardcode base = ? done via #282]*** ? P4 ? P5 Launch-Report flip ? P6 GPU parity). ? P1's v6 migration coordinates with the M8 v6 cost-ledger migration ? author the version bump once.

22 6. ? ****Ops-Agent codified (#291)**** ? `deploy/gcp/create\_gpu\_vm.sh` (the one-command VM helper) always bakes the Ops-Agent startup-script; provision.sh grants the APIs+SA roles. PROVEN active on a fresh VM 2026-06-21. ([[gcp-ops-agent-enable]])

23 7. ****[owner] Set the ship-gate target (#172)**** ? needs corpus growth.

24 8. ****[owner-side GPU] R-series: promote-or-reject `R2` tolF1 as the gate**** (the ablation) ? needs GPU-extracted golden features.

25

26 ****P2 ? later / gated / deferred****

27 9. Rally-quality ****P3/P4**** (person-fusion LOVO, audio cue) ? needs GPU golden features.

28 9b. ? ****DONE (#396):**** the R4 density fix (finding B / P2b in `RALLY\_DETECTION\_FIXES\_PLAN.md`) is ****promoted to default-ON****. Golden-set A/B at the ****SERVED**** windowing showed tolF1 0.353?0.364 (?tolF1 +0.010, no regression), so `inactivity\_temporal\_density` default was flipped to True. (Landed flag-gated default-OFF in #394.) Fixed the sparse-tracking quirk where one fast sample after a detection gap held the rally tail open.

29 10. ****M0.4 scorer swap**** ? TemporalMaxer (per the 2026-06-17 external review).

30 11. ****Multisport**** (table-tennis first, then pickleball).

31 12. Serving ****M8/M9**** ? per-video cost ledger, edge auth, DPA/consent (within `COMPUTE\_DECOUPLED\_SERVING`; owner/legal-gated).

32 13. ****WASB-C3**** weight provenance (gates *sharing* a trained model) ? legal.

33 14. ****Rename repo off "badminton" (owner, BACKLOG).**

34 15. Owner housekeeping: retire the shared ****D0 droplet**** + ****D0 token****; collector ****md5**** export (collector repo).

35 16. ****PMF field validation**** (owner-side, business track).

36

37 ****Doc-hygiene (tracked in `docs/DOC\_STATUS.md` \$4):**** ? #1 per-video reconcile + ? #2 this refresh; ****pending:**** field/PMF/Roora ? archives + a data-layer `docs/` dir (owner boundary confirm); confirm the COMPLETED top-level stubs; verify `HOSTING\_PLAN`/`UI\_PROPOSAL` currency.

38

39 ---

40

41 **## Owned-model & rally-quality track ? detail (2026-06-13 plan; still valid)**

42

43 ****Updated:**** 2026-06-13 (****rally-detection quality track****: fusion + experiment-harness design docs merged,

44 owner-gated ? see the dedicated section below, the desktop-agent pick-up). Prior: 2026-06-10 (audit?M0?Gemini-battery session; later same day the ****UI-alpha track opened****:

45 ADR-009 ratified ? KhelSutra Guru / khelsutra.guru, private `khelsutra` repo, local-first delivery ? see

46 `docs/UI\_ALPHA\_EXECUTION\_PLAN.md`; ****PR-1** async jobs (#16) MERGED 2026-06-11, PR #87**;

47 ****PR-2** upload/download MERGED 2026-06-11, PR #99** ? live-E2E-verified (3-part chunked upload MD5 byte-identity +

48 compile?browser-download); next: PR-3 identity. Same day: the ****8?9 quality sweep #90?#98 landed**** ? ruff/mypy/

49 coverage-floor CI lanes, SQLite conn-close fix, config unknown-key warning, hybrid-twin collapse into

50 `trajectory\_hybrid.py`, WSL tilde+abspath live-path fixes; suite 388 green; LOVO 0.626 re-verified exact).

6. user (2026-07-03T16:40:17.301Z)

1 # Roadmap ? from annotated data to a self-improving, offline rally segmenter

2

3 This document is the ****narrative thread****: how the indexer evolves from an

4 AI-API-dependent prototype into a self-contained tool that learns rally detection

5 from its own commercially-licensed, annotated video corpus. It explains *how we

6 got here*, *why each step matters*, and *how it shapes the final tool*. Decisions

7 are recorded as ADRs in [DECISIONS.md](archives/decisions/DECISIONS.md); the scoring

8 mechanics live in EVALUATION.md.

```

9
10     > ? **Phase 4 update:** the authoritative owned-model path is now
11     > [`OWNED_MODEL_IMPLEMENTATION_PLAN.md`](OWNED_MODEL_IMPLEMENTATION_PLAN.md) ? a learned
**per-frame TAL head**
12     > (ActionFormer/ASFormer over WASB trajectory features), not the candidate-window
classifier sketched below.
13     > **Audio is parked, not "gated on golden labels":** the E1 probe and a follow-up
foreground-audio test both came back
14     > weak (~0.6 AUC), so it is de-prioritized even though labels now exist (see the plan's
Gen-1 note).
15
16     ## The thesis
17
18     The indexer detects badminton rallies in long recordings. Its strongest mode
19     (`yolo_hybrid`) is a two-stage pipeline: cheap **permissive person-detection
20     player-motion** filtering (Stage 1, local ? torchvision Faster R-CNN + ByteTrack
21     tracking) followed by **Gemini** semantic refinement (Stage 2, paid API). That
22     works, but it costs money, needs the network, and never improves on its own.
23
24     > *Naming note: the `yolo_hybrid` registry key is retained for back-compat, but
25     > AGPL YOLOv8 was removed and replaced with a BSD/MIT-licensed torchvision +
26     > ByteTrack stack ? see **ADR-005**. "YOLO" in this doc means the local
27     > player-motion gating stage, not the Ultralytics library.
28
29     A sibling project, **sports-data-collector**, curates sports videos with
30     per-rally `[start, end)` annotations and a commercial-license flag. The thesis of
31     this roadmap: **use that annotated, commercially-usable data to (1) measure the
32     pipeline objectively and (2) train an offline replacement for the paid Gemini
33     stage** ? turning the tool into one that improves as more data is curated, runs
34     air-gapped, and generalizes across sports.
35
36     Why it matters: it removes the per-run API cost and network dependency, gives us
37     a defensible quality metric instead of eyeballing, and creates a flywheel ?
38     more curated matches ? a better offline segmenter ? better highlights.
39
40     ## The four phases
41
42     ### Phase 1 ? Bridge the repos (DONE)
43     A sport-agnostic `curate export` in the collector emits the indexer's `start,end`
44     ground-truth CSVs plus a `manifest.json` pairing each CSV with its video, gated on
45     the `is_commercial` flag. ? *Why: one clean producer/consumer contract that works
46     for all five supported sports (badminton/tennis/cricket/pickleball/table-tennis) alike.
47     See **ADR-002**.
48
49     ### Phase 2 ? Acquire full-resolution video (DONE)
50     The corpus was 256x144 (the original crawl used `worstvideo`). The `fetch` command

```

7. user (2026-07-03T16:40:17.615Z)

```

1     # Golden-labeled video registry
2
3     **Purpose.** The single source of truth for every **golden** (human rally-boundary-
labeled) video ? what
4     it is, where the source proxy lives, how many rallies, and whether it's ingested into
the eval corpus.
5     Use this to **isolate the golden videos into a dedicated location** and to keep the
growing golden set
6     auditable. Updated whenever a new video is labeled + ingested.
7
8     > **This is the GROWING GOLDEN-finalized set** in the dual-eval-set discipline (see
9     > [`QUALITY_ITERATIONS.md`](../QUALITY_ITERATIONS.md), the 2026-06-16 capstone + `eval-
dual-set-regression`):
10     > a **RAW eval set** (currently-bad videos) + this **golden-finalized eval set**
(promoted when at-par) +
11     > a separate **golden TRAINING set**. Every quality launch must not regress on either

```

```

eval set.
12     > Privacy-safe, video-free *shipping* of these =
[GOLDEN_REGRESSION_FIXTURES.md](../GOLDEN_REGRESSION_FIXTURES.md).
13
14     ## Contracts / conventions
15     - **Labels:** rally-annotator CSV
('rally_number,start_time,end_time,ending_reason,sport,shots_count`,
16     decimal seconds) ? ingested by `backend/eval/gt_loader.py` (reads
`start_time`/`end_time`).
17     - **Alignment is mandatory before trusting any score.** The label timebase MUST match
the windowed proxy
18     (same fps + frame count). Verify with `ffprobe` (the `largetest_doubles` near-miss is
why) ? ideally
19     annotate the *exact* proxy the detector ran on (then `video_id` = MD5 matches and
alignment is free).
20     - **`video_id`** = MD5 of the proxy (`backend/utils/hashing.py::compute_video_id`); it
keys the WASB
21     trajectory cache, the Gemini reference rows, and the golden features.
22     - **Ingestion** = copy the trajectory + labels to durable paths, write
`output/<name>_golden_features.json`
23     (`gts` + candidates), append a row to `output/human_lovo_manifest.json`.
(`scratch/at_par.py` scores a
24     single video vs Gemini + golden; `backend/eval/rally_seg_eval.py` runs the whole
corpus.)
25     - **Guardrails:** owned footage only; **minors excluded** ; labels never derived from a
paid LLM (Gemini is
26     inference/reference only). Commercial-clean.
27
28     ## Ingested golden corpus ? `output/human_lovo_manifest.json` (n = 15)
29
30     | # | name | sport | rallies | fps | source proxy | video_id | labeled |
31     |---|---|---|---|---|---|---|---|
32     | 1 | adarsh_avi_singles | badminton (S) | 96 | 59.94 | (original 6) | ? | pre-06-16 |
33     | 2 | kushagra_singles | badminton (S) | 32 | 59.94 | (original 6) | ? | pre-06-16 |
34     | 3 | largetest_doubles | badminton (D) | 49 | 29.97 | (original 6) | ? | pre-06-16 |
35     | 4 | gbaaddy | badminton | 48 | 59.94 | (original 6) | ? | pre-06-16 |
36     | 5 | testlarge_short | badminton | 6 | 29.97 | (original 6) | ? | pre-06-16 |
37     | 6 | mahadevpura_singles | badminton (S) | 31 | 29.97 | (original 6) | ? | pre-06-16 |
38     | 7 | mahadevpura_2 | badminton (multicourt) | 40 | 59.94 | `?/Roora Request - June
15/[Done] Video 3 ?/mahadevpura-2_proxy.mp4` | `038e0e0915a2` | 2026-06-16 |
39     | 8 | GX010128 | badminton (multicourt) | 52 | 59.94 | `?/Golden Label Videos Request -
June 15/[Done] Video 13 ?/GX010128.MP4` | `db84f3e65318` | 2026-06-16 |
40     | 9 | mahadevpura_1 | badminton (S, continuous) | 10 | 59.94 | `?/[Done] Video 14 -
Badminton/mahadevpura-1.MP4` | `38bef75d78cb` | 2026-06-17 |
41
42     ## Labeled, ingestion in progress (WASB trajectory extracting)
43
44     | name | sport | rallies | fps | source proxy | status |
45     |---|---|---|---|---|---|
46     | GX030094 | badminton | 41 | **29.97** | `?/Golden Label ?/[Done] Video 6 -
Badminton/GX030094_proxy.mp4` | **ingested** (in `human_lovo_manifest.json`) |
47     | Badminton BXH 2 | badminton (multicourt) | 44 | 59.94 | `?/[Done] Video 10 - Badminton
Multicourt/Badminton BXH 2_proxy.mp4` | **ingested** (in `human_lovo_manifest.json`) |
48
49     ## Source folder & not-yet-labeled
50     - **Annotation root:** `G:\My Drive\Sharing\Annotation Videos\Golden Label Videos
Request - June 15`

```

8. assistant (2026-07-03T16:40:19.815Z)

Now let me search for specific mappings between section 7, 8, and 10 to identify concrete redundancies and drift risks:

9. user (2026-07-03T16:40:21.305Z)

13:The corpus is now large enough to move from "grow labels first" into alpha-readiness work, but the 15-video headline is not itself a launch gate pass. `[verified]` The golden manifest has 15 badminton videos; `[analysis]` the local manifest needed path rebasing before manifest-driven commands ran; `[analysis]` only 6 of 15 feature JSONs had the full fusion schema; `[analysis]` the n=15 quality baseline is stale and must be intentionally refreshed before it can gate anything.

```
33:| D3 | **The 15-video corpus is now a measurement asset, not a free launch pass.** |
`data_pipeline/GOLDEN_VIDEOS.md`; local eval runs, 2026-07-03 | Rebase/portability, feature
completeness, baseline refresh, and gate ratification come before quality claims. |
53:- `[analysis]` There are 15 `output/*_golden_features.json` files, but only 6 carry full
shuttle/person/audio candidate features. The newer 9 are windowing-only schema and are skipped
by fusion/person eval.
54:- `[analysis]` `python -m backend.eval.fusion_compare --features-dir output` loaded only 6
videos and measured person-fusion `Delta F1 = -0.021`; not promotable.
58:- `[analysis]` `python -m backend.eval.ablation --features-dir output --granularity group`
currently fails at import due to a circular import between `backend/eval/ablation.py` and
`backend/eval/ablation_pdf.py`.
62:### 4.1 Alpha lane: make the product usable
66:### 4.2 Quality lane: make the 15-video corpus gate-ready
70:### 4.3 Flywheel lane: make tester feedback useful
79:| Stale local manifests create false confidence | Manifest-driven tools skipped or failed
until paths were rebased | Make the corpus manifest portable and add a path/fixture smoke check
|
80:| 15-video corpus is only partially feature-complete | Fusion/person/audio eval is still n=6
today | Regenerate or upconvert full-fusion features for all 15 before fusion decisions |
91:## 7. Deliverables & progress tracker
99:| A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke check
that fails loudly when labels or trajectories are missing | `badminton-highlight-indexer` | A0 |
No | TODO | - |
103:| A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation on
the n=15 corpus | `badminton-highlight-indexer` | A2, A4 | No | TODO | - |
104:| A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so
fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data
availability | TODO | - |
105:| A8 | Re-run `fusion_compare`, group ablation, and served contrast after A7; explicitly
promote/reject person fusion, Gate A served-default, and any other R-series default flips |
`badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | TODO | - |
112:## 8. Concrete next PR sequence
117:4. **Baseline PR:** intentional n=15 nightly baseline refresh and heuristic floor refresh.
118:5. **Ablation/fusion PRs:** fix ablation import, complete 15-video features, rerun
promotion/rejection decisions.
127:5. **Engineering:** Should n=15 feature regeneration target full fusion schema for every
clip, or should the eval loaders support mixed schemas with explicit capability tags?
129:## 10. Definition of done
135:- [ ] R2/tolF1 gate is promoted or rejected with a recorded ablation result.
160:python -m backend.eval.fusion_compare --features-dir output
164:python -m backend.eval.ablation --features-dir output --granularity group
```

10. assistant (2026-07-03T16:40:23.140Z)

Let me get the full sections 7, 8, and 10 to do detailed comparison:

11. user (2026-07-03T16:40:23.937Z)

```
91      ## 7. Deliverables & progress tracker
92
93      Legend: TODO = not started, IN PROGRESS = actively being worked, DONE =
shipped/verified, GATED = blocked on owner/legal/platform decision. One PR per row unless the
row explicitly names a cross-repo batch.
94
95      | ID | Deliverable | Repo / owner | Depends on | Gated? | Status | PR |
96      |----|-----|-----|-----|-----|-----|----|
97      | A0 | Create this alpha readiness tracker and index it in docs | `badminton-highlight-
indexer` | - | No | IN PROGRESS | #451 |
98      | A1 | Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check,
```

```

edge auth ON, upload -> process -> jobs -> compile -> download verified | `khelsutra` +
`badminton-highlight-indexer` | A0 | Owner spend / CF config | TODO | - |
99      | A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke
check that fails loudly when labels or trajectories are missing | `badminton-highlight-indexer`
| A0 | No | TODO | - |
100     | A3 | Promote the 15-video source-video/MD5 records into the collector/vault path;
reconcile collector's six-video source manifest with the 15-video eval corpus | `sports-data-
collector` + vault | A2 | Owner upload/storage scope | TODO | - |
101     | A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact
command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2
| Reviewer sign-off | TODO | - |
102     | A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic_lovo_n6`
floor for future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | TODO | - |
103     | A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/tolF1 ablation
on the n=15 corpus | `badminton-highlight-indexer` | A2, A4 | No | TODO | - |
104     | A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so
fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data
availability | TODO | - |
105     | A8 | Re-run `fusion_compare`, group ablation, and served contrast after A7; explicitly
promote/reject person fusion, Gate A served-default, and any other R-series default flips |
`badminton-highlight-indexer` | A6, A7 | Reviewer sign-off | TODO | - |
106     | A9 | Run Gen-0/TemporalMaxer shadow training against the refreshed n=15 floor; produce
a non-serving artifact bundle and gate verdict | `badminton-highlight-indexer` | A5, A8 | Owner
approval for M0.4 scope | TODO | - |
107     | A10 | Decide and record the alpha ship-gate target: metric, corpus slice, minimum
threshold, significance rule, and what claims are allowed | Owner + `badminton-highlight-
indexer` | A4, A5, A8 | Owner decision | GATED | - |
108     | A11 | Resolve or explicitly defer WASB C3 provenance for alpha messaging; if deferred,
document "no commercial model sharing" copy guardrail | Owner/legal | A10 | Legal/provenance |
GATED | - |
109     | A12 | Reopen or defer P10/P11 corpus flywheel: `PATCH /api/segments`, collector
promotion, `train|reeval` job, and delta-F1 surface | `khelsutra` + `badminton-highlight-
indexer` + collector | A1, A3 | Owner D13 reopen | GATED | - |
110     | A13 | Alpha launch review: runbook, privacy/consent copy, known limits,
rollback/disable steps, and owner GO/NO-GO | Cross-repo | A1, A4, A10, A11 | Owner launch
decision | TODO | - |
111
112     ## 8. Concrete next PR sequence
113
114     1. **Docs PR:** land this tracker.
115     2. **Corpus portability PR:** rebase/portable manifest tooling plus missing-file smoke
test.
116     3. **Hosted alpha PR/batch:** health/auth/upload/process/job/compile/download
verification and runbook updates.
117     4. **Baseline PR:** intentional n=15 nightly baseline refresh and heuristic floor
refresh.
118     5. **Ablation/fusion PRs:** fix ablation import, complete 15-video features, rerun
promotion/rejection decisions.
119     6. **Launch decision PR/docs update:** record ship-gate target, C3 status, alpha copy
limits, and launch checklist.
120
121     ## 9. Open questions
122
123     1. **Owner:** Is the first alpha allowed to use manual/offline corpus promotion, or
should P10/P11 be reopened before any tester wave?
124     2. **Owner/platform:** Which hosted compute target is the next stand-up: Cloud Run GPU,
GCE L4 VM, or another approved GPU host?
125     3. **Owner/legal:** What is the acceptable alpha wording while WASB C3 remains
unresolved?
126     4. **Engineering:** Should the portable corpus manifest live as a generated artifact, a
checked-in redacted manifest, or both?
127     5. **Engineering:** Should n=15 feature regeneration target full fusion schema for every
clip, or should the eval loaders support mixed schemas with explicit capability tags?
128
129     ## 10. Definition of done

```

```
130
131     - [ ] Invited tester can complete the hosted record/upload -> reel download CUJ without
SSH.
132     - [ ] Alpha service is gated behind the chosen auth posture and has a documented
disable/rollback path.
133     - [ ] n=15 corpus is portable enough that manifest-driven quality commands do not rely
on stale local absolute paths.
134     - [ ] n=15 nightly baseline and heuristic floor are refreshed with recorded commands.
135     - [ ] R2/tolF1 gate is promoted or rejected with a recorded ablation result.
136     - [ ] Fusion/person/Gate-A served-default decisions are explicitly promoted or rejected
on the 15-video corpus.
137     - [ ] Ship-gate target and allowed alpha claims are recorded.
138     - [ ] WASB C3 status is recorded and reflected in alpha messaging.
139     - [ ] P10/P11 are either reopened with a committed sequence or explicitly deferred for
alpha-plus.
140     - [ ] Owner records GO/NO-GO for the first alpha wave.
141
142     ## 11. References
143
144     Internal docs:
145
146     - `docs/NEXT_STEPS.md`
147     - `docs/ROADMAP.md`
148     - `docs/data_pipeline/GOLDEN_VIDEOS.md`
149     - `docs/R2_EVAL_METRIC_DESIGN.md`
150     - `docs/RALLY_DETECTION_FIXES_PLAN.md`
```

12. assistant (2026-07-03T16:40:26.109Z)

Now I have the complete picture. Let me analyze for simplification issues. I'll identify concrete redundancies, divergences, and maintenance costs:

13. user (2026-07-03T16:40:27.141Z)

```
cd07339 docs: add alpha launch readiness tracker
e4914d1 Surface validation failure reasons
f0814d5 Normalize upload filenames before extension checks
e585965 Fix ffmpeg concat clip paths (#449)
ac6184b docs: archive the completed 2026-07-02 workspace audit umbrella (#446)
```

14. assistant (2026-07-03T16:40:29.680Z)

Now let me map the redundancies systematically:

15. user (2026-07-03T16:40:29.933Z)

```
60     ## 4. Launch strategy
61
62     ### 4.1 Alpha lane: make the product usable
63
64     The alpha lane is successful when an invited tester can open the app, authenticate,
upload a video, watch the job progress, pick rallies, compile a reel, and download it. This lane
should avoid model-quality promises beyond the measured state and should default to the most
reliable serving path.
65
66     ### 4.2 Quality lane: make the 15-video corpus gate-ready
67
68     The quality lane is successful when the 15-video corpus is portable, reproducible,
feature-complete, and backed by refreshed baselines. Only then should R2 tolF1,
Gen-0/TemporalMaxer, or other model/promotion decisions be ratified.
69
70     ### 4.3 Flywheel lane: make tester feedback useful
71
72     The flywheel lane is successful when user corrections can flow into the collector/vault
```

path with consent/provenance and a visible re-eval delta. This is valuable for alpha learning, but it should not block the first controlled alpha if manual promotion can bridge the gap.

73

74 ## 5. Risks and guardrails

75

76 | Risk | Why it matters | Guardrail |

77 |---|---|---|

78 | Launch activity implies model launch approval | The engine docs explicitly say rally-detection product launch is NO-GO until ship target and WASB C3 clear | Keep all alpha copy scoped to invite-only processing, not owned-model claims |

79 | Stale local manifests create false confidence | Manifest-driven tools skipped or failed until paths were rebased | Make the corpus manifest portable and add a path/fixture smoke check |

80 | 15-video corpus is only partially feature-complete | Fusion/person/audio eval is still n=6 today | Regenerate or upconvert full-fusion features for all 15 before fusion decisions |

81 | Baseline refresh accidentally rubber-stamps a regression | Nightly runner correctly reported stale baselines | Refresh baselines only after reviewer sign-off and record the command/output |

82 | P10/P11 expands scope before hosted alpha is usable | Corpus flywheel is valuable but can consume the alpha launch window | Keep P10/P11 behind explicit owner reopen unless alpha lane is already stable |

83 | WASB checkpoint provenance remains unresolved | Blocks commercial sharing of a trained model regardless of F1 | Treat C3 as a hard model-release gate; do not hide it inside product launch tasks |

84

85 ## 6. Honest limits

86

87 Completing this doc's alpha lane does ****not**** prove the owned model is ready, that WASB weights are commercially clean, that the product can serve unbounded public traffic, that multi-sport is ready, or that every tester correction automatically improves the model.

88

89 Completing the quality lane does ****not**** by itself deploy a service. It only makes the measurement surface honest enough for a launch decision.

90

91 ## 7. Deliverables & progress tracker

92

93 Legend: TODO = not started, IN PROGRESS = actively being worked, DONE = shipped/verified, GATED = blocked on owner/legal/platform decision. One PR per row unless the row explicitly names a cross-repo batch.

94

95 | ID | Deliverable | Repo / owner | Depends on | Gated? | Status | PR |

96 |----|-----|-----|-----|-----|-----|----

97 | A0 | Create this alpha readiness tracker and index it in docs | `badminton-highlight-indexer` | - | No | IN PROGRESS | #451 |

98 | A1 | Stand up alpha serving endpoint: Cloud Run/GPU or approved host, health check, edge auth ON, upload -> process -> jobs -> compile -> download verified | `khelesutra` + `badminton-highlight-indexer` | A0 | Owner spend / CF config | TODO | - |

99 | A2 | Make the n=15 golden manifest portable/reproducible; add a path/corpus smoke check that fails loudly when labels or trajectories are missing | `badminton-highlight-indexer` | A0 | No | TODO | - |

100 | A3 | Promote the 15-video source-video/MD5 records into the collector/vault path; reconcile collector's six-video source manifest with the 15-video eval corpus | `sports-data-collector` + vault | A2 | Owner upload/storage scope | TODO | - |

101 | A4 | Refresh the n=15 nightly regression baseline intentionally and record the exact command/output; keep stale-baseline warnings until reviewed | `badminton-highlight-indexer` | A2 | Reviewer sign-off | TODO | - |

102 | A5 | Recompute the heuristic n=15 floor and replace the stale `heuristic\_lovo\_n6` floor for future Gen-0 comparisons | `badminton-highlight-indexer` | A2 | No | TODO | - |

103 | A6 | Fix `backend.eval.ablation` CLI circular import, then run the R2/toIF1 ablation on the n=15 corpus | `badminton-highlight-indexer` | A2, A4 | No | TODO | - |

104 | A7 | Regenerate or upconvert full-fusion feature JSONs for the newer 9 videos so fusion/person/audio analyses run over all 15 | `badminton-highlight-indexer` | A2 | GPU/data availability | TODO | - |

Based on my analysis of the diff, I've identified the key simplification issues. Let me compile the findings:

```
```json
[
 {
 "file": "docs/ALPHA_LAUNCH_READINESS.md",
 "line": "112-119",
 "summary": "§8 'Concrete next PR sequence' steps 2, 4, 5 restate § 7 rows A2, A4-A5, A6-A7 verbatim but without dependency/gating info, creating drift risk: if A2 depends on A0 but step 2 omits this, a reader executing §8 linearly may miss blocking. Derive §8 from §7's dependency graph or delete it to avoid maintenance of two execution orders.",
 "failure_scenario": "Owner executing §8 step 2 before A0 completes causes false confidence that 'corpus portability' is unblocked; they discover the gating issue only after starting the work."
 },
 {
 "file": "docs/ALPHA_LAUNCH_READINESS.md",
 "line": "129-140",
 "summary": "§10 'Definition of done' items 2-10 restate §7 table rows verbatim (A1?item 1, A2?item 3, A4?item 4, A6?item 5, A8?item 6, A10?item 7, A11?item 8, A12?item 9, A13?item 10) with no added semantics. §7's 'Status' column is the authoritative progress source; §10 creates a parallel checkbox list that must be kept in sync or diverges.",
 "failure_scenario": "A row like A6 moves to DONE in §7 but §10 checkbox remains unchecked, creating confusion about actual alpha readiness; the checkbox list becomes cargo-cult if it mirrors the table instead of defining new acceptance criteria."
 },
 {
 "file": "docs/ALPHA_LAUNCH_READINESS.md",
 "line": "74-83",
 "summary": "§5 'Risks and guardrails' row 'Stale local manifests create false confidence' restates the problem solved by §7 A2 ('Make the n=15 golden manifest portable/reproducible') and §10 item 3 ('n=15 corpus is portable enough') in identical terms. The guardrail and the deliverable are the same work; consolidate into a single 'corpus portability' requirement or explain why the risk framing adds value beyond the task definition.",
 "failure_scenario": "A reviewer asks 'is A2 done?' and a maintainer checks §5's guardrail (not updated) or §10's checkbox (not marked DONE); they interpret stale guardrails as ongoing risk despite A2 shipping, leading to false confidence that the risk is not resolved."
 },
 {
 "file": "docs/ALPHA_LAUNCH_READINESS.md",
 "line": "74-83 and 129-140",
 "summary": "§5 guardrail 'Fusion/person/audio eval is still n=6 today' ? 'Regenerate or upconvert full-fusion features for all 15 before fusion decisions' exactly mirrors §7 A7 and §10 item 6. The risk table and the checklist both assert this is a gate for 'fusion decisions,' but §7 shows A7 and A8 (the actual promotion/rejection step) are separate, creating ambiguity: does A7 alone unblock fusion, or does A8 decide whether to apply fusion at all?",
 "failure_scenario": "After A7 lands (feature regeneration), a team member assumes fusion/Gate A can be 'promoted or rejected' (§7 A8 language) without realizing A8 depends on A6 (ablation import fix) and A7 together; they run fusion_compare on the n=15 features but ablation still fails, wasting review cycles."
 },
 {
 "file": "docs/ALPHA_LAUNCH_READINESS.md",
 "line": "62-72 and 95-110",
 "summary": "§4 'Launch strategy' defines three lanes (Alpha, Quality, Flywheel) as conceptual success criteria, but §7 deliverables A0-A13 do not map 1:1 to lanes: Alpha lane tasks (A1 serving + A2-A8 quality readiness prep) are mixed with gated architectural decisions (A10, A11, A12). The lane framing and the sequential dependency table serve different mental models; readers conflate 'we need Quality lane done before Alpha' with 'A2-A8 before A1,' but A1 and A4-A8 can parallel (A1 gated on owner/CF, not A2-A3). Clarify lane?rows mapping or merge §4 into §7 as prose context.",
 "failure_scenario": "An engineer interprets §4.2 'Quality lane is successful when the 15-video corpus is portable, reproducible, feature-complete, and backed by refreshed baselines' as a hard blocker for Alpha (§4.1). They wait for A7 + A8 to complete before starting A1,
```



missing the fact that A1 and quality work are parallel and only A2 (portability) is sequential to both."

```
 },
 {
 "file": "docs/ALPHA_LAUNCH_READINESS.md",
 "line": "33 and 66-68",
 "summary": "§2 decision D3 states 'The 15-video corpus is now a measurement asset, not a free launch pass' and implies portability/completeness/baseline refresh are preconditions. §4.2 repeats this verbatim: 'when the 15-video corpus is portable, reproducible, feature-complete, and backed by refreshed baselines.' Both are true, but the repetition suggests D3 is just a reformulation of the Quality lane goal rather than a distinct locked decision. Either move D3 into the preamble as framing, or clarify what locks D3 (the decision) beyond §4.2 (the criterion).",
 "failure_scenario": "A new reader sees D3, then later sees §4.2 with nearly identical wording, assumes one is stale or redundant, and either ignores D3 as cargo-cult or second-guesses whether the Quality lane is mandatory for alpha ship."
 }
]
```